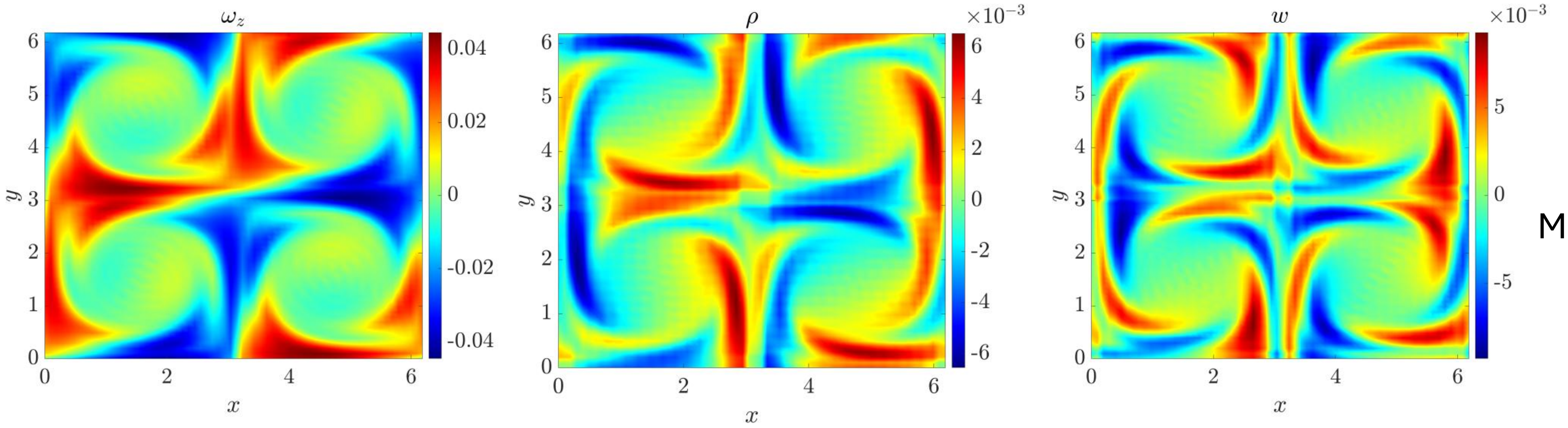
Zigzag instability of columnar Taylor–Green vortices

Fr	k	σ	**The growth rate from matrix-free code**
1.000	2	0.2108 + 0.0000i	0.2117
0.500	3	0.2251 + 0.0000i	0.2256
0.250	5	0.2422 + 0.0000i	0.2423
0.125	9	0.1998 + 0.0000i	0.1999

Table 1. The most unstable mode at each Fr considered.

Guo J, Taylor JR, Zhou Q. Zigzag instability of columnar Taylor–Green vortices in a strongly stratified fluid. Journal of Fluid Mechanics. 2024 997:A34.



Matrix-free code

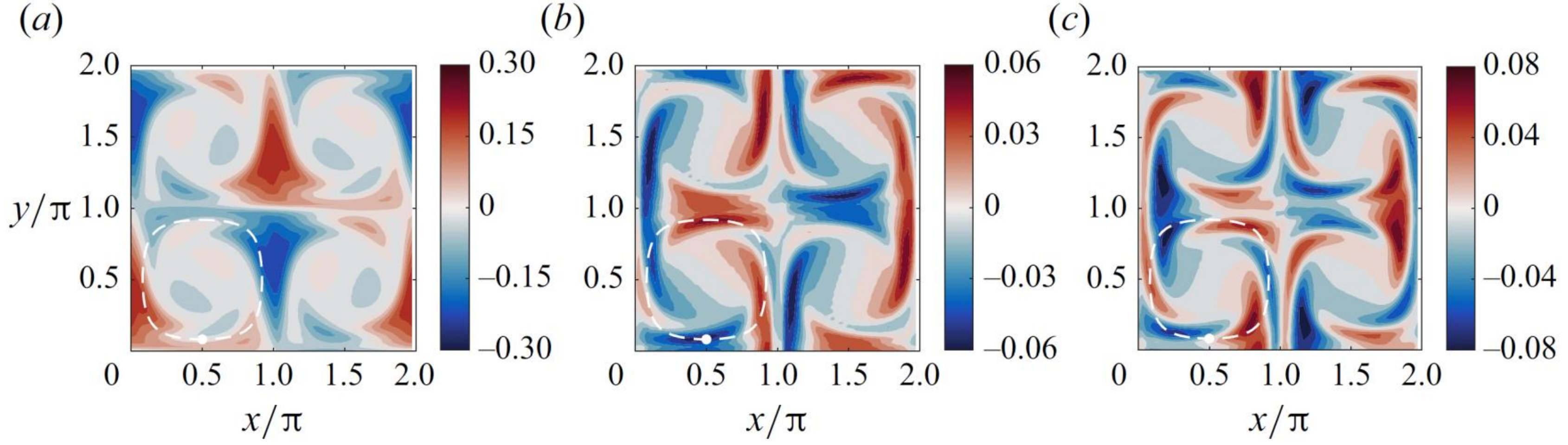
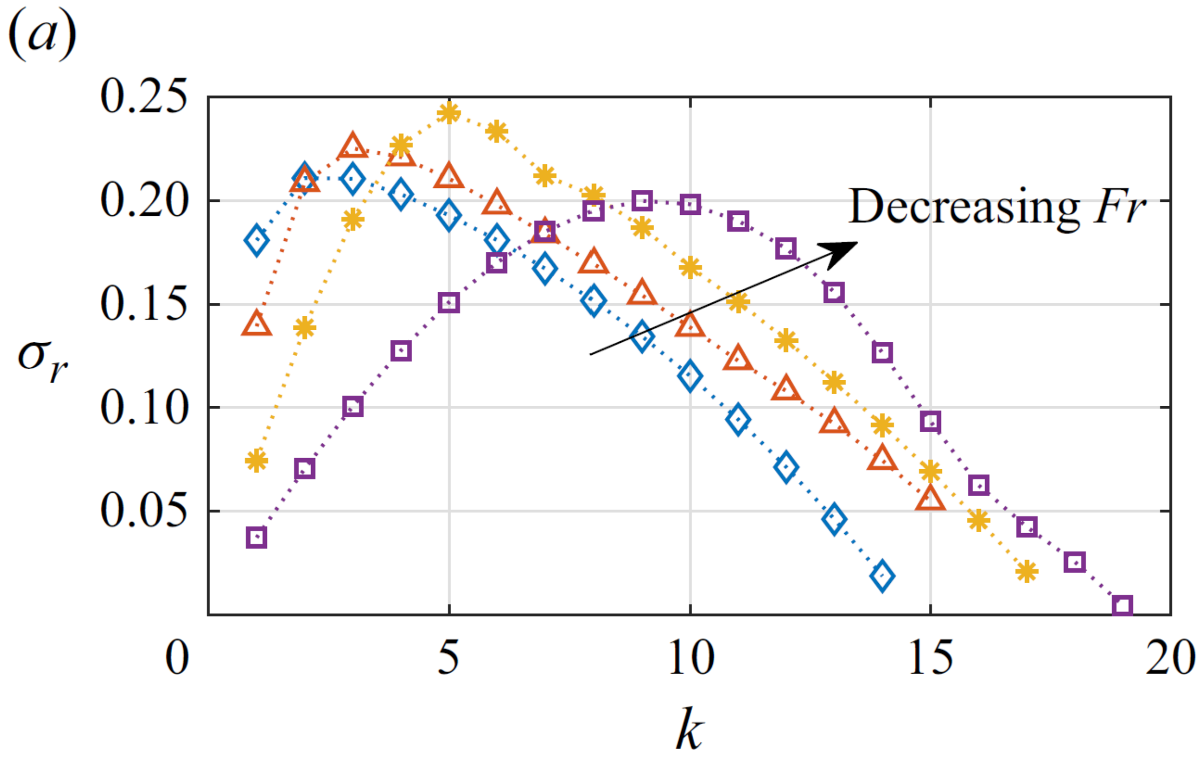


Figure 4. Horizontal transect of perturbation (*a*) ω_z at $z = z_0$, (*b*) ρ at $z = z_+$, and (*c*) w at $z = z_+$, for the eigenmode with $k = 3$ at $Fr = 0.5$. The heights z_0 and z_+ are specified in figure 5. The amplitude of the eigenmode is normalized such that the kinetic energy associated with the horizontal perturbation velocities (u, v) constitutes 1 % of the kinetic energy in the base flow (U, V). The dashed line indicates the streamline corresponding to $\psi = 0.25$, and the solid dot indicates the starting point of a particle trajectory considered in figure 6.

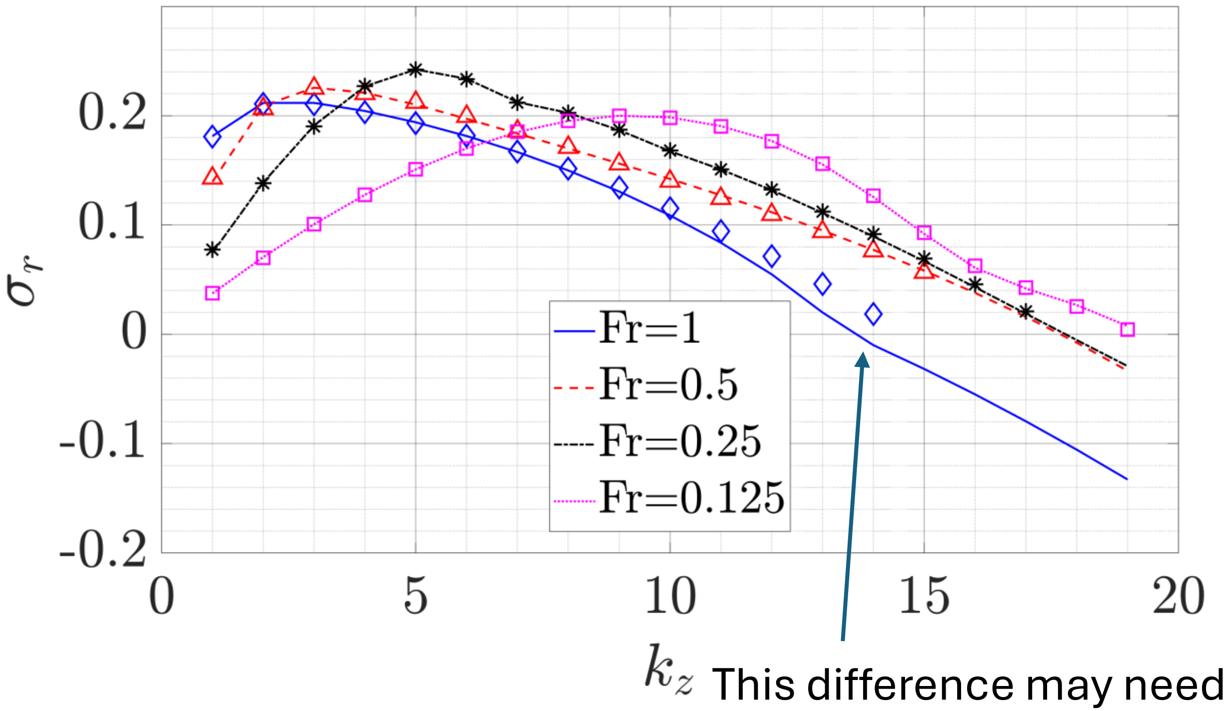
Guo J, Taylor JR, Zhou Q. Zigzag instability of columnar Taylor–Green vortices in a strongly stratified fluid. Journal of Fluid Mechanics. 2024 997:A34.

Figure 2(a) of Guo J, Taylor JR, Zhou Q. Zigzag instability of columnar Taylor–Green vortices in a strongly stratified fluid. Journal of Fluid Mechanics. 2024 997:A34.



Lines: matrix-free code
Markers: from Guo, Taylor & Zhou (2024)
Nx=Ny=64

Select the stationary modes to compare growth rate

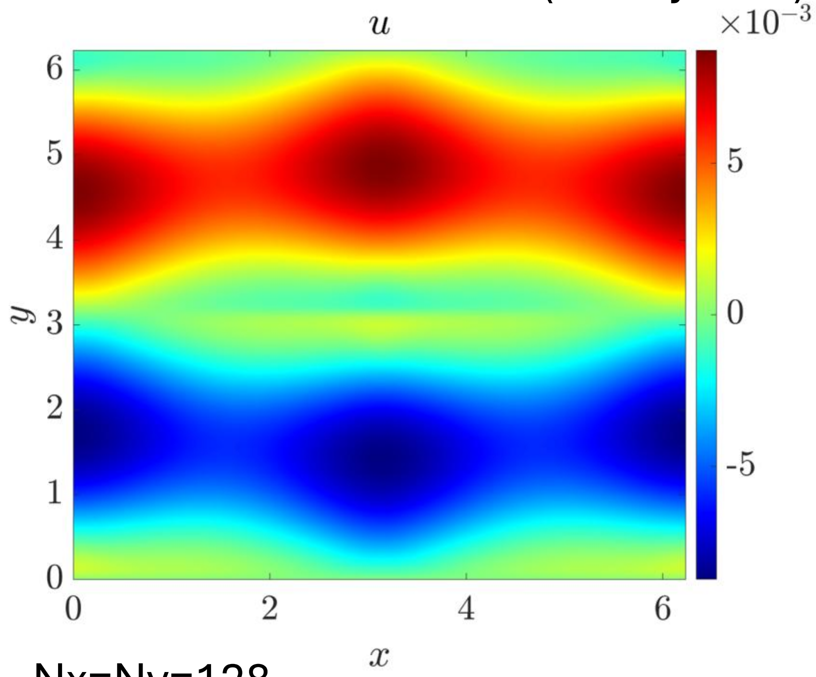


This difference may need higher resolution to resolve

Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7

A0_viscous

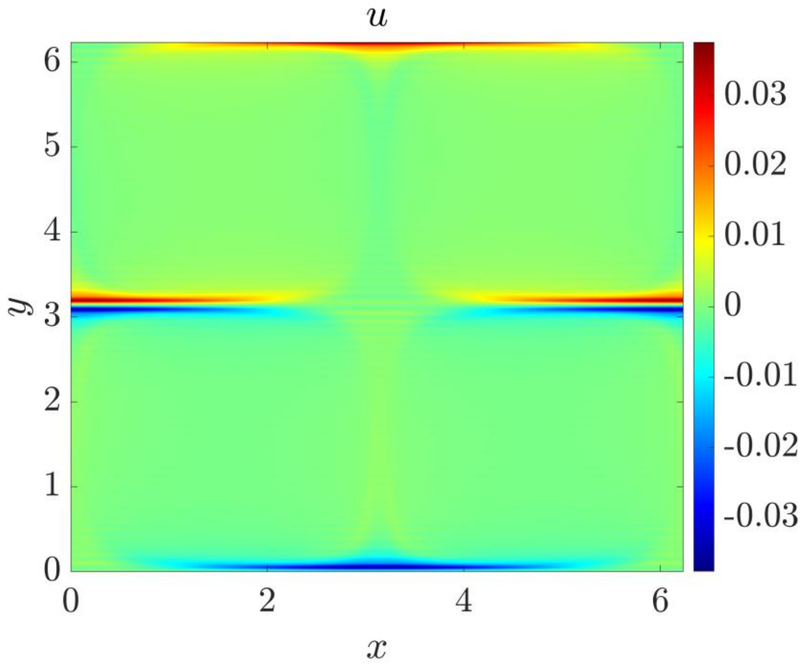
lambda= 0.1409 - 0.5932i (Nx=Ny=64)
lambda= 0.1337 - 0.5925i (Nx=Ny=128)



Nx=Ny=128

A0_adj_viscous

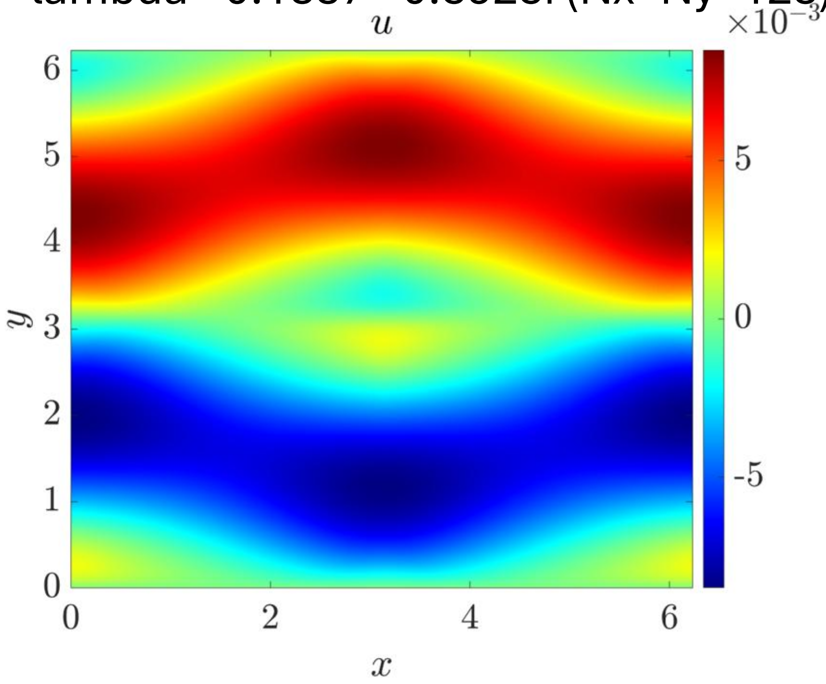
lambda= 0.1409 + 0.5932i (Nx=Ny=64)
lambda=0.1337 - 0.5925i (Nx=Ny=128)



Nx=Ny=128

A_full, with kz=0.0001

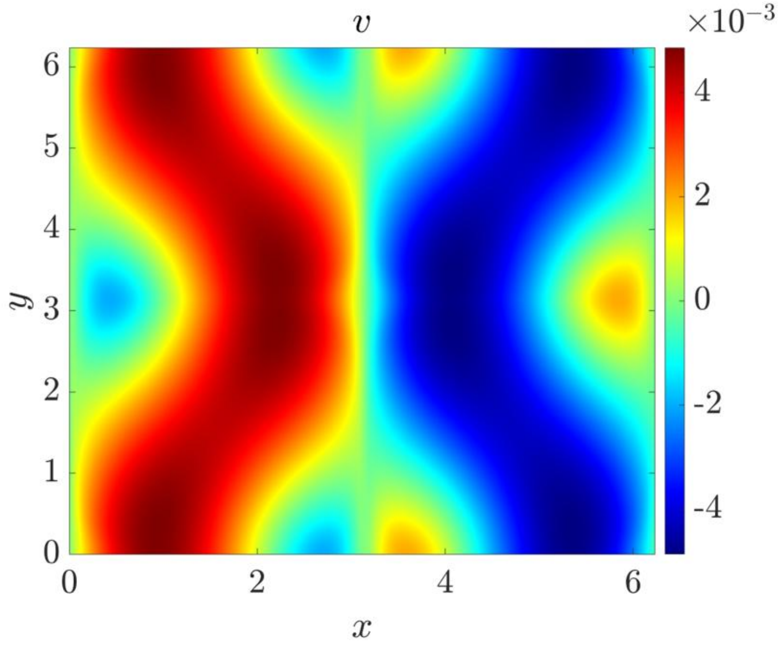
lambda= 0.1409 - 0.5932i (Nx=Ny=64)
lambda= 0.1337 - 0.5925i (Nx=Ny=128)



Nx=Ny=128

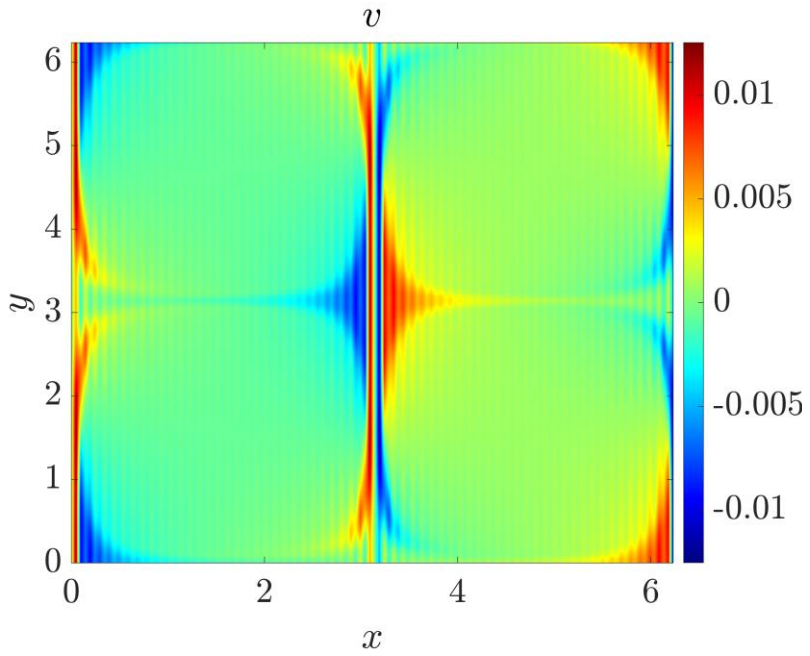
Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7

A0_viscous



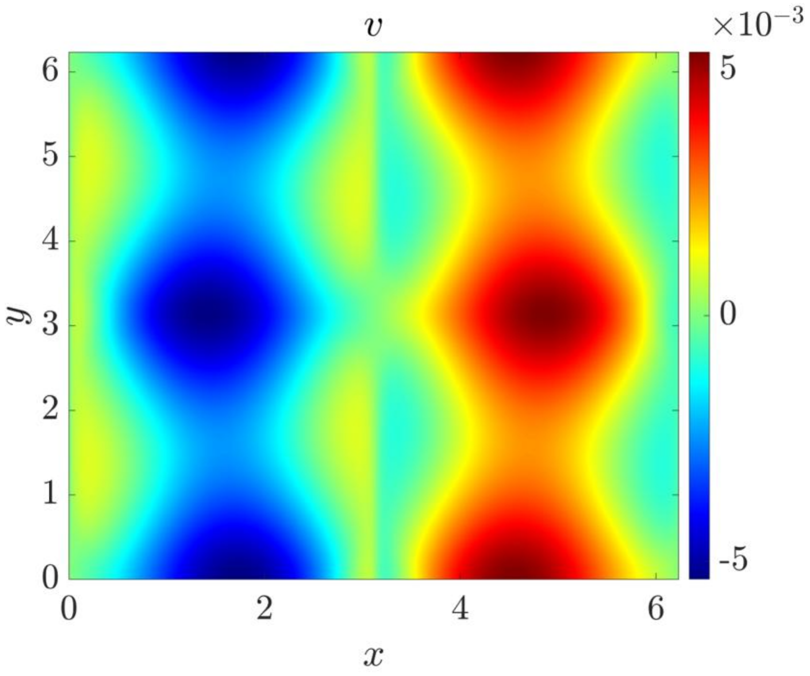
Nx=Ny=128

A0_adj_viscous



Nx=Ny=128

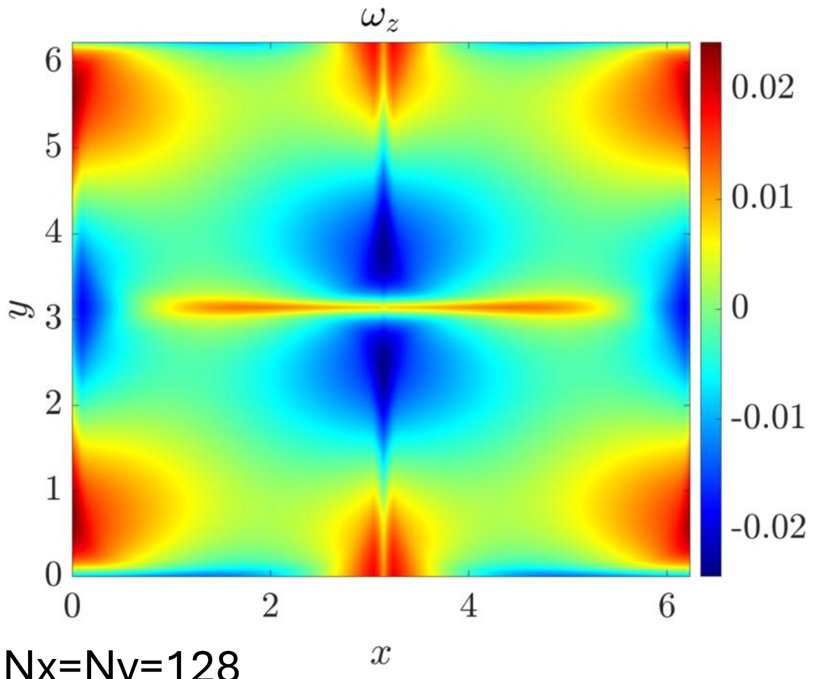
A_full, with kz=0.0001



Nx=Ny=128

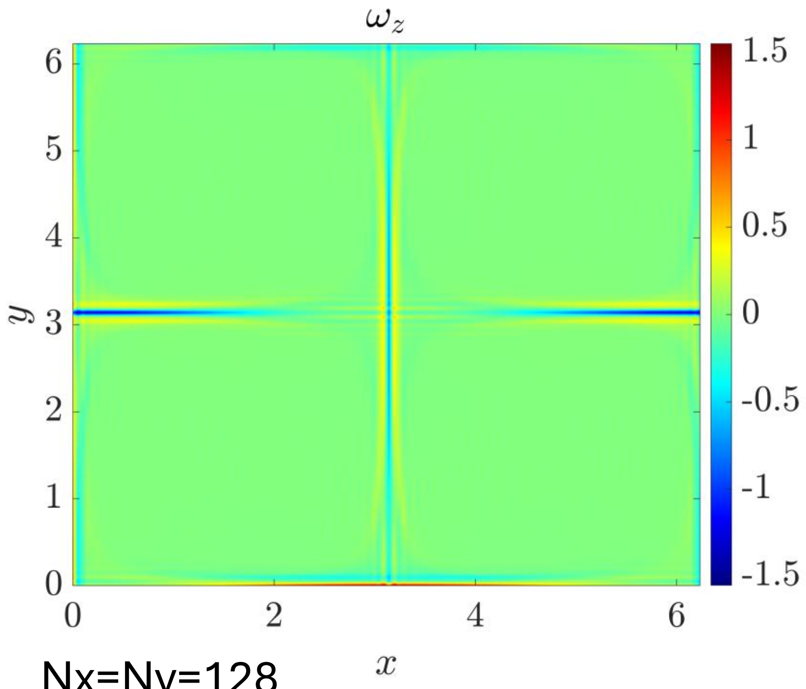
Comparing eigenvectors associated with A0_viscous, A0_adj_viscous, and A_full operators: Fr=0.125, Re=1600, Pr=0.7

A0_viscous



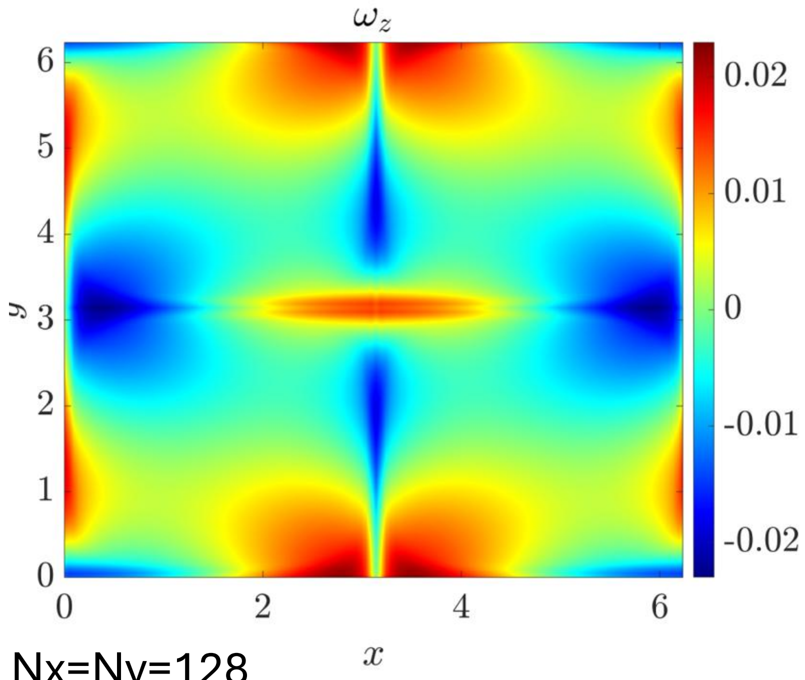
Nx=Ny=128

A0_adj_viscous



Nx=Ny=128

A_full, with kz=0.0001



Nx=Ny=128